

8.3 Networks

The main focus of advanced network services is in Japan and Korea. While most of the world is yet to see 3G devices, the Far East is buzzing with murmurs of 4G services. What lies in the near future for countries such as the US, the UK, and India is already old news in the East. While many enjoy streaming video services with advanced features and bandwidth of the Mbps variety, we in India are still languishing with WAP-enabled browsing and pathetically slow GPRS speeds. Will this change? Yes, we will get better services, but perhaps will never catch up with the crazy East.

However, there are still some software and technologies that give us hope.

WiMAX

Though far from being implemented in India, WiMAX does offer a chance of achieving a “connected” India. Since WiMAX is much superior to existing WiFi technologies, and offers better speeds and an enhanced range, there’s no doubting the fact that it might be our only hope. If India deployed a nationwide WiMAX network, which is possible given the enormous coverage offered by WiMAX, we could truly realise the dream of networking even the tiniest of villages.

WiMAX offers a range of up to 10 km per cell site, and speeds of up to 40 Mbps per channel. This is more bandwidth than cellular networks or ISPs will ever need. The equipment is expensive however, and it will be some time before we, and the service providers as well, will be able to afford such equipment.

UMA

UMA stands for Unlicensed Mobile Access. No, it’s not a hacking technique or anything illegal. UMA quite simply uses GSM and GPRS services over an unlicensed frequency spectrum such as Bluetooth or WiFi.